

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Movie Review Topic Discovery Using LSA

Sentiment analysis in R

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THE QUESTION

Star ratings show whether viewers liked a film—not why

Star ratings tell us whether viewers liked a film, not why. The problem is to automatically discover the latent themes present in a large collection of movie reviews and to measure the sentiment associated with each theme, so that the reasons behind audience opinion become visible without manual reading.

Objectives:

- Clean and structure raw review text for analysis.
- Identify the most informative words using term frequency and tf-idf.
- Discover k latent topics using LSA and label them from their highest-loading terms.
- Score review sentiment using the AFINN, Bing and NRC dictionaries and compare them.
- Combine topics and sentiment to show which themes viewers praise or criticize.
- Validate the sentiment results against the labels provided in the dataset.



Discover

Discover recurring themes across many reviews.



Measure

Measure the sentiment associated with each theme, so that the reasons behind audience opinion become visible without manual reading.

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THE SCENARIO

Consider the analytics team of a streaming platform or production house. Each release attracts thousands of reviews, and the team needs quick, evidence-based answers to three questions.

Streaming teams need to turn review volume into answers

01 What are viewers talking about most (acting, plot, direction, music, visuals)?

02 Which themes drive praise and which drive complaints?

03 What should be acted on in content decisions, marketing and recommendations?

The pipeline in this report is intended to answer these questions at scale.



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THE DATASET

The IMDb dataset pairs 50,000 reviews with balanced labels

The proposed dataset is the IMDb Large Movie Review Dataset (Maas et al., 2011), also available on Kaggle as ‘IMDB Dataset of 50K Movie Reviews’.



25,000

positive reviews

25,000

negative reviews

Size — 50,000 movie reviews. Class balance — 25,000 positive and 25,000 negative reviews. Columns — Review text and a sentiment label. Use of labels — Allow validation of the sentiment analysis. Practical note — A random sample of 5,000 to 10,000 reviews can be used if the full set is slow in R. Alternatives — Rotten Tomatoes reviews; the smaller Cornell polarity dataset (2,000 reviews).

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THE PIPELINE

The workflow has six stages, each tied to a unit of the course syllabus.

Six stages take reviews from raw text to validated insight

01

Explore and clean text

Review length, class balance and summary statistics are examined first. The text is then cleaned: converted to lowercase, stripped of HTML tags such as
, punctuation and numbers, and stopwords are removed. Stemming or lemmatization is optional.

02

Tokenize; calculate word frequency and tf-idf

Using unnest_tokens(), each review is split into words, one per row. Word frequencies show the most common terms. The bind_tf_idf() function then weights terms so that words distinctive to a group of reviews score higher than words common everywhere, in line with Zipf's law, which explains why a few words dominate raw counts.

03

Build a document-term matrix

Reviews form the rows and terms the columns, with tf-idf weights as the values. This matrix is the input to LSA.

04

Discover and label LSA topics

LSA applies Singular Value Decomposition (SVD) to the matrix, described in Section 6. The number of topics k is chosen with a scree plot or cumulative variance explained. Each topic is interpreted and named from its highest-loading terms.

05

Score sentiment with three dictionaries

Reviews are scored with the three dictionaries described in Section 7. The most common positive and negative words are listed, word clouds are drawn, and average sentiment is calculated per topic.

06

Compare with dataset labels; report in R Markdown

Lexicon-based sentiment is compared with the true labels using accuracy and a confusion matrix. Optionally, a logistic regression classifier is trained on the review features and its predictions are explained with LIME. All steps are documented in R Markdown.

Keynote point: validation and c
make the workflow repeatable.

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THE METHOD

LSA uses SVD to reveal latent themes in review language

The workflow has six stages, each tied to a unit of the course syllabus.

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Reviews form the rows and terms the columns, with tf-idf weights as the values. This matrix is the input to LSA.

$$A \approx U_k \Sigma_k V_k^T$$

LSA applies Singular Value Decomposition (SVD) to the matrix, described in Section 6. The number of topics k is chosen with a scree plot or cumulative variance explained. Each topic is interpreted and named from its highest-loading terms.

A	SVD	latent themes
term × document	$U_k \Sigma_k V_k^T$	high-loading terms topic labels

Reviews are scored with the three dictionaries described in Section 7. The most common positive and negative words are listed, word clouds are drawn, and average sentiment is calculated per topic.

Lexicon-based sentiment is compared with the true labels using accuracy and a confusion matrix. Optionally, a logistic regression classifier is trained on the review features and its predictions are explained with LIME. All steps are documented in R Markdown.

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SENTIMENT ANALYSIS

Three sentiment dictionaries offer different lenses



AFINN

AFINN — Assigns each word a score from –5 to +5 for strength of feeling.



Bing

Bing — Labels words as simply positive or negative.



NRC

NRC — Positive and negative labels plus emotions such as joy, anger and fear.

Comparing the three dictionaries shows how sensitive the conclusions are to the choice of lexicon. Sentiment is then aggregated per review and per topic, which connects the themes found by LSA with the opinions expressed about them.

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EXPECTED OUTPUTS

Expected outputs connect topics with audience opinion

A set of interpretable topics, for example acting and performances, plot and story, horror and thrills, family and comedy.

Interpretable themes

A set of interpretable topics, for example acting and performances, plot and story, horror and thrills, family and comedy.

Topic views

Visual outputs: word clouds, bar charts of top terms and a topic-versus-sentiment chart.

Opinion links

A sentiment profile for each topic, showing which themes tend to be positive or negative.

Validation

A measured accuracy for the sentiment approach against the labels in the dataset.

This report is design-stage: present these as expected outcomes, not actual measured findings.

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LIMITS & NEXT STEPS

Lexicons and topic choices create known limits—and clear next steps



Model choices

LSA loadings can be negative, so topics are harder to interpret than those from LDA. The choice of k affects the topics and requires judgment.



Lexicon blind spots

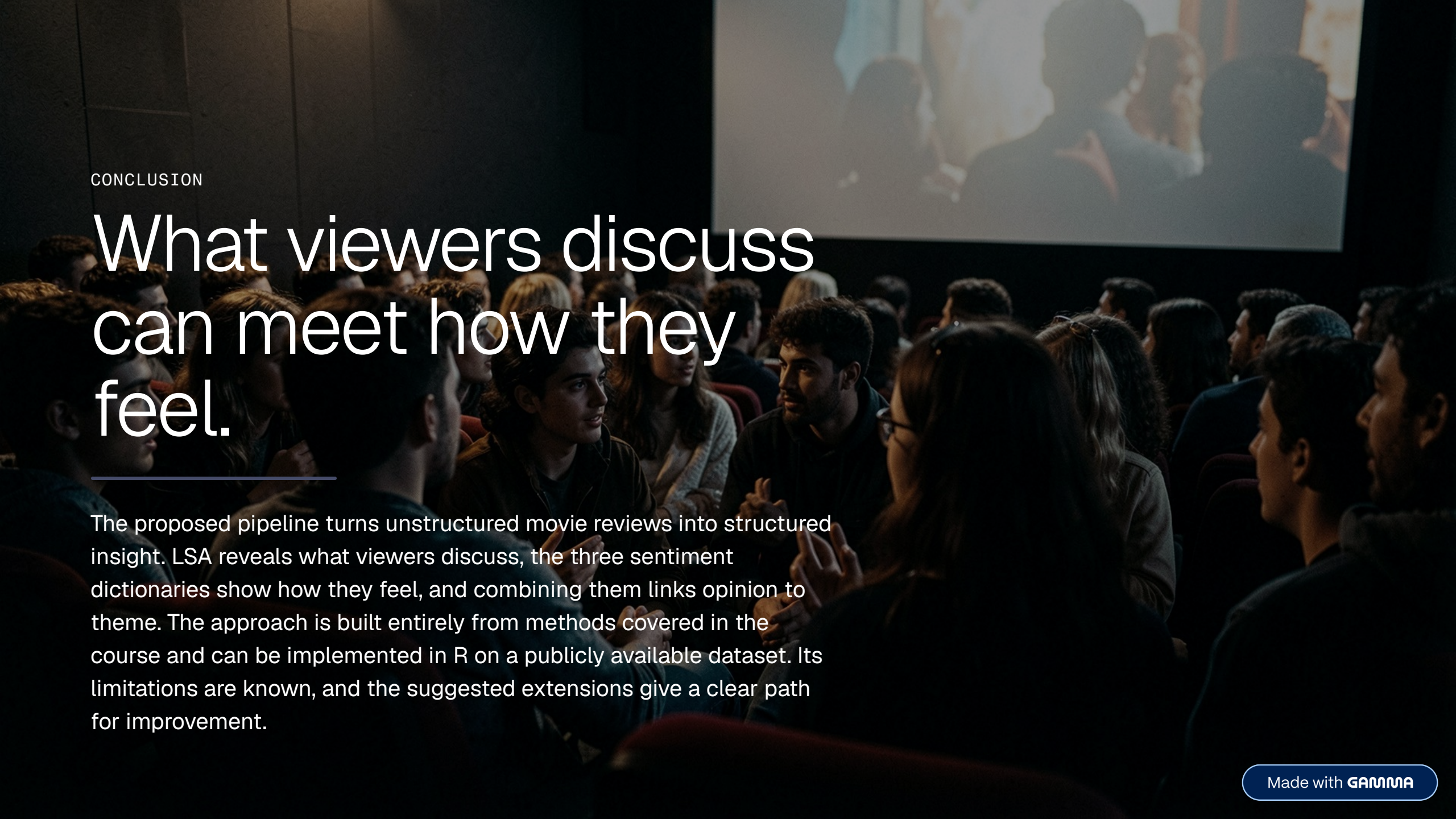
Dictionary-based sentiment misses negation (‘not good’) and sarcasm. Domain-specific words can mislead: a word like ‘kill’ is negative in a lexicon but neutral in a thriller review.



Next steps

LDA topic modeling for clearer topics. Transformer-based sentiment models. Aspect-level sentiment for each topic.

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CONCLUSION

What viewers discuss can meet how they feel.

The proposed pipeline turns unstructured movie reviews into structured insight. LSA reveals what viewers discuss, the three sentiment dictionaries show how they feel, and combining them links opinion to theme. The approach is built entirely from methods covered in the course and can be implemented in R on a publicly available dataset. Its limitations are known, and the suggested extensions give a clear path for improvement.

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